Milestone 2 – Alpha: Progress Document (Sentinel Protocol)

1. Project information

This document describes the Alpha build of Sentinel Protocol, a solo, Blueprint-only combat arena made in Unreal Engine 5.8.

Item	Detail
Project title	Sentinel Protocol
Developer	Naz Yagmur (individual project)
Engine	Unreal Engine 5.8, Blueprint-only
Project file	NAZ_3023.uproject
Map to open	Content/SentinelProtocol/Maps/SentinelProtocol_MainMap
GitHub	github.com/nazyg/NAZ_3023
Demo video	youtu.be/BCBwcWK8qs4

Controls: W, A, S, D to move; mouse to look; left mouse button to fire; T applies 10 damage to the player (test key); R restarts the round.

Windows build: unzip the build folder and run NAZ_3023.exe. Press Alt + F4 to quit. The project folder is also submitted, and the same version is on GitHub.

2. Alpha build overview

The Alpha build is one complete, repeatable round: the player fights three Sentinel drones in the arena built for Milestone 1 and wins by destroying all of them.

1. The player spawns at the PlayerStart with 100 health, a crosshair and a HUD showing health, score and time.
2. Sentinel drones detect the player with AI Perception and shoot at them while they have a clear line of sight.
3. The player shoots back with a hitscan weapon (20 damage per hit). A drone has 60 health, so it takes 3 hits.
4. An electric floor panel and an Energy Cell add risk and recovery on the way.
5. Each destroyed drone adds 1 to the score. After the third drone, the timer stops and ROUND COMPLETE appears.
6. If health reaches 0, SYSTEM FAILURE appears and the player respawns after 3 seconds without reloading the level. Score and timer keep running.
7. Pressing R restarts the round with score and timer reset.

3. Alpha requirements status

All ten Alpha requirements are implemented and working in the current build.

Requirement	Status	How it is met
Playable arena	Done	Blockout from Milestone 1: entry, upper balcony, lower hall, Control Room, main floor
Reliable combat	Done	Hitscan weapon (line trace from the camera), fire-rate limit, crosshair
Consistent damage and health	Done	One reusable health component (AC_Health) and one damage path (Apply Damage) for every source
Clear death state	Done	SYSTEM FAILURE message, input disabled
Respawn without restarting	Done	GameMode respawns the player at the PlayerStart after 3 seconds
Functional AI opponent	Done	BP_SentinelDrone: AI Perception, turns to the player, fires on a timer, can be destroyed
Initial match state	Done	Score, timer, win condition (3 drones), ROUND COMPLETE, R to restart
Environmental hazard	Done	BP_ElectricFloor: switches on and off every 2 seconds, damages the player while active
First pickup	Done	BP_EnergyCell: heals the player by 30, then disappears
Repeatable stability	Done	Three full rounds and repeated death and respawn tested without crashes (section 8)

4. Core systems

Every combat system shares one health component and one damage path, so the player, the drones and the target dummies behave the same way.

4.1 Health component (AC_Health)

AC_Health is an Actor Component with the variables MaxHealth, Health and bIsDead. It has two Event Dispatchers: OnHealthChanged (Health, MaxHealth) and OnDeath. The function ApplyDamageAmount ignores damage if the owner is already dead, subtracts the damage, clamps Health between 0 and MaxHealth, broadcasts OnHealthChanged, and broadcasts OnDeath once Health reaches 0. Heal works the same way in the other direction. The player has 100 health and a drone has 60.

4.2 HUD (WBP_HUD)

The HUD has a health bar, a crosshair, a score text, a timer text, a SYSTEM FAILURE text and a ROUND COMPLETE text. The character creates the HUD in BeginPlay, then binds to OnHealthChanged to update the bar and to OnDeath to show the death message. Score, time and the round message read their values from the GameMode through UMG property bindings. The character removes the HUD from the screen in EndPlay, so a respawn never creates a second HUD.

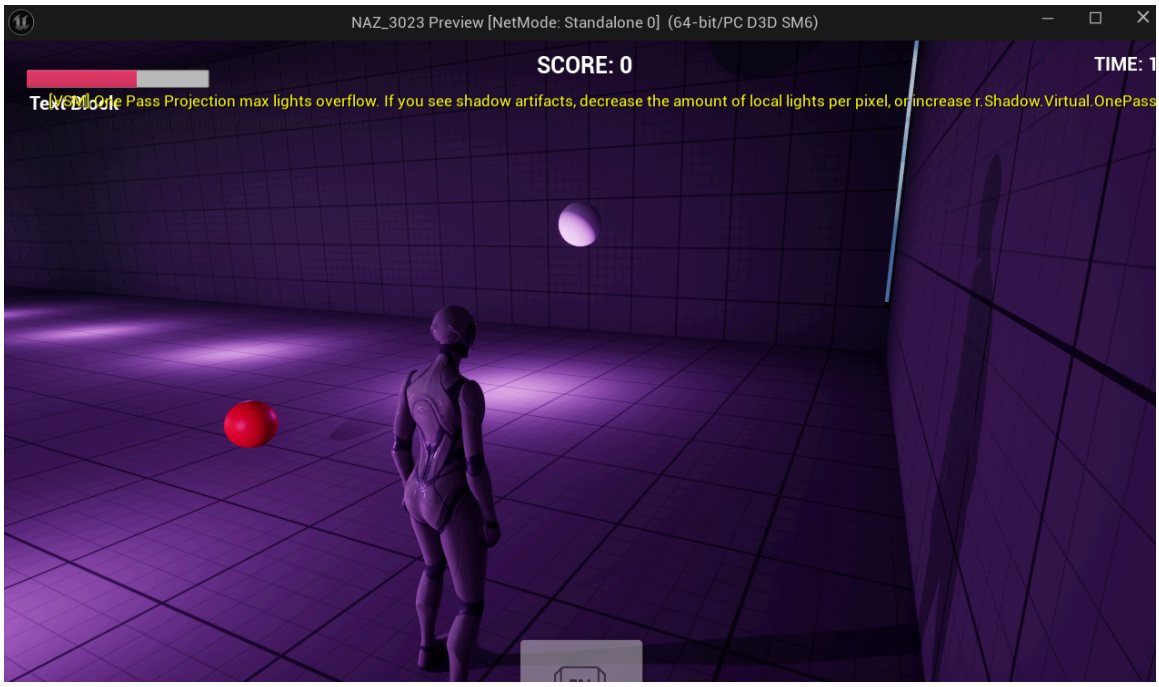


Figure 1 – HUD in play: health bar (top left), score (top centre) and timer (top right).

4.3 Hitscan weapon

The Enhanced Input action IA_Fire (left mouse button, hold to fire) checks a CanFire flag, then runs two things: a cooldown (CanFire off, Delay of 0.3 seconds, CanFire on) and a Line Trace By Channel on the Visibility channel. The trace starts at the camera location and ends at the camera location plus the camera forward vector times 5000. If it hits an actor, Apply Damage sends 20 damage to that actor. A red debug line shows each shot, and the crosshair marks where the trace goes.

4.4 Death and respawn

When OnDeath fires, the character disables player input, shows SYSTEM FAILURE and asks BP_SentinelGameMode to respawn it. The GameMode waits 3 seconds, destroys the old pawn and calls Restart Player, which spawns a new character at the PlayerStart with full health. The level is never reloaded, so the score and the timer continue.

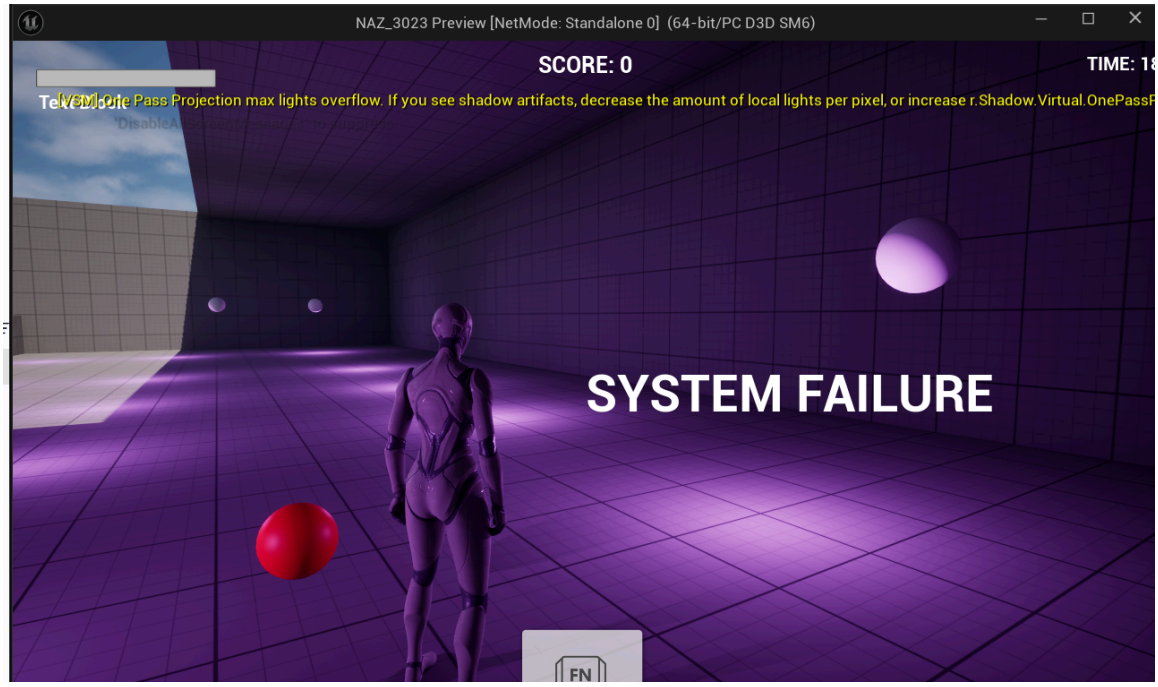


Figure 2 – Death state: the player has no health left and SYSTEM FAILURE is shown before the respawn.

4.5 Sentinel Drone (AI opponent)

BP_SentinelDrone is a stationary turret with a sphere mesh, an AC_Health component (60 health) and an AI Perception component with an AI Sight config (sight radius 1200, lose-sight radius 1400, 180 degree half angle, Detect Neutrals on).

- **Detect:** On Target Perception Updated casts the sensed actor to the player character, then sets or clears the variable TargetActor.
- **Fire:** a looping timer calls TryFire every FireInterval seconds. TryFire checks that TargetActor is valid, turns the drone to face it, and traces a line to it. The drone only fires if nothing blocks the line, so walls protect the player. The shot applies FireDamage through Apply Damage and draws a red debug line.
- **Die:** Event AnyDamage feeds ApplyDamageAmount. On death the drone tells the GameMode that it was killed, then destroys itself.

FireDamage, FireInterval and the sight radius are variables, which made it easy to tune the difficulty during testing.

4.6 Match state (BP_SentinelGameMode)

The GameMode keeps Score, TargetKills (3), TimeSeconds and bRoundOver. The custom event DroneKilled adds 1 to Score and sets bRoundOver when Score reaches TargetKills. A looping 1-second timer adds to TimeSeconds until the round is over. The R key runs the console command RestartLevel, which resets everything for a new round.

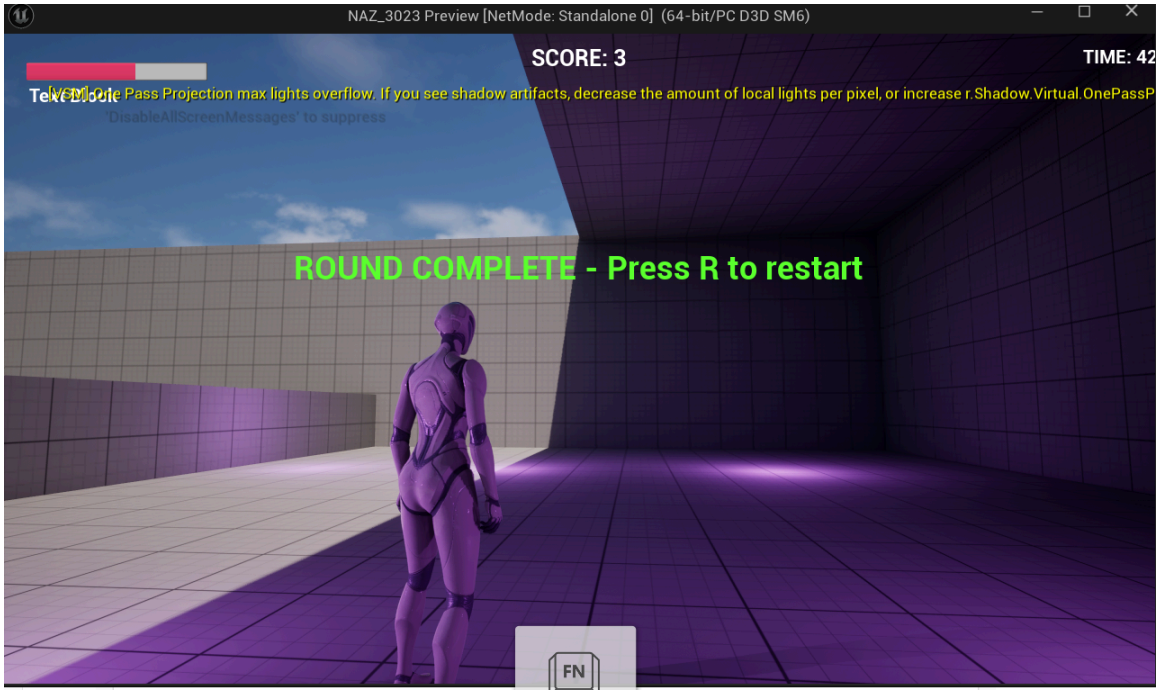


Figure 3 – Round complete: score 3, timer stopped, prompt to press R to restart.

5. Hazard, pickup and door

These three actors add environmental risk, recovery and a chokepoint to the arena.

Actor	What it does	How it works
BP_ElectricFloor	Floor panel that is dangerous for 2 seconds, then safe for 2 seconds	A looping timer flips the bActive flag every 2 seconds and swaps the material (dark when off, glowing red when on). A second 0.5-second timer reads the overlapping player from a box collision and applies DamagePerTick (10) through Apply Damage while bActive is true
BP_EnergyCell	Pickup that heals 30 health	A sphere collision detects the player on overlap, calls the Heal function of AC_Health with HealAmount (30), then destroys the pickup
BP_SlidingDoor	Door at the Control Room entrance	Reused from Milestone 1: a box trigger plays a Timeline that slides the door up, and a 1.5-second delay closes it again after the player leaves

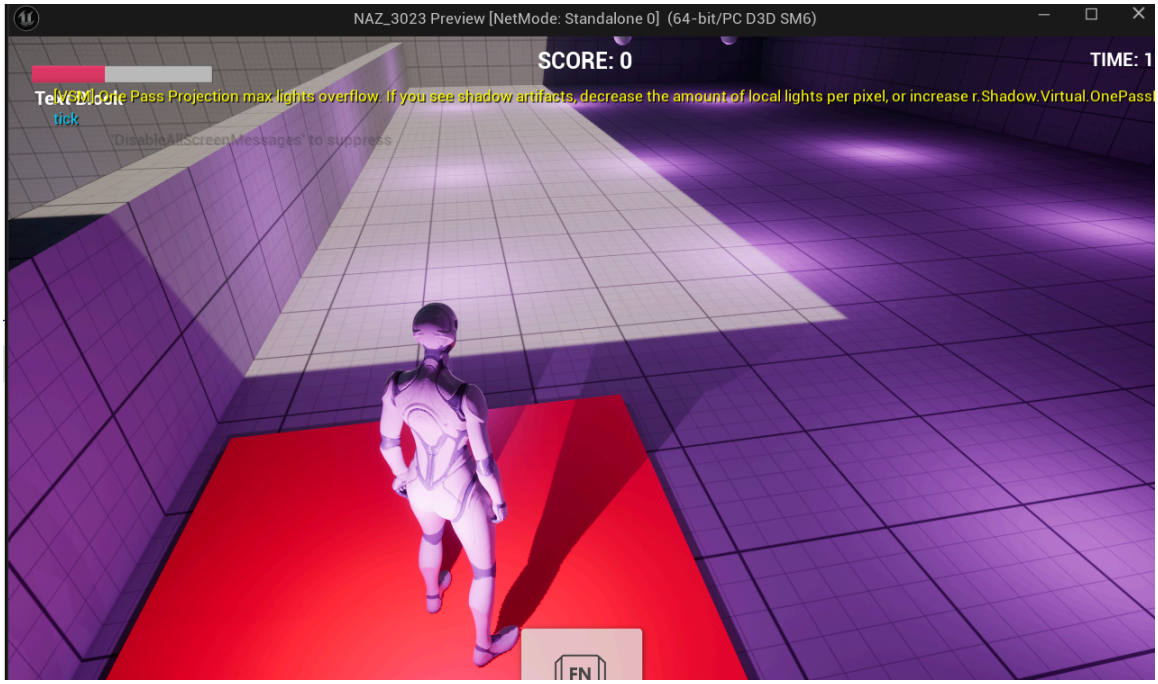
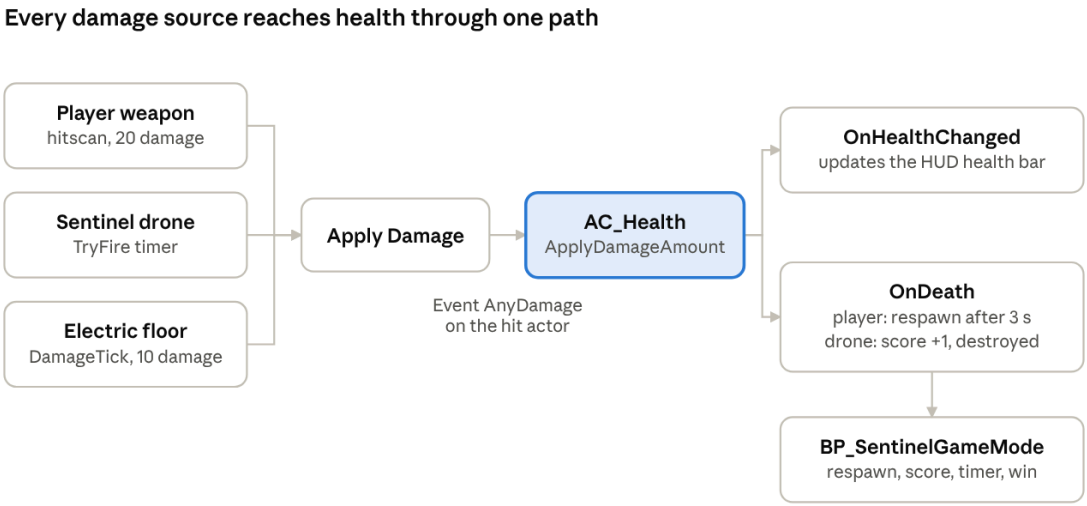


Figure 4 – Electric floor in its active (red) state with the player standing on it.

Because the hazard uses the same Apply Damage path as weapons, damage from the floor updates the HUD, triggers death and respawns the player without any extra code.

6. Architecture: one damage path



damage flow · 3 sources, 1 path

The weapon, the drone and the electric floor all call Apply Damage. The hit actor's Event AnyDamage passes the value to AC_Health, whose two dispatchers drive the HUD, the respawn and the score.

7. Problems found and how they were fixed

Most bugs were wrong pin connections, found by playtesting and by reading the Blueprint graphs.

Problem	Cause	Fix
Health bar did not update	The OnHealthChanged dispatcher input MaxHealth was a Boolean instead of a Float	Changed the input type to Float and reconnected the getter
Blueprint error "Accessed None" on the HUD	The HUDRef variable was never set from the Create Widget return value	Connected Create Widget to Set HUDRef before Add to Viewport
Shots went in the wrong direction	The trace end point used only forward vector times range, without adding the camera location	Added a Vector + Vector node: camera location plus (forward vector times FireRange)
Target dummy took hits but never died	Event AnyDamage and the OnDeath binding were missing in the dummy Blueprint	Added Event AnyDamage to ApplyDamageAmount, and OnDeath to Destroy Actor
Electric floor changed colour but did no damage	The DamageTick event was not connected to its Branch, and Base Damage was left at 0	Connected the event and wired DamagePerTick into Apply Damage
Energy Cell healed but stayed in the level	Destroy Actor was not connected after Heal	Connected Heal to Destroy Actor
Drone killed the player too quickly	The sight radius was long and damage was high	Reduced the sight radius, damage and fire rate (all are variables), so the player can outrange the drone

8. Testing results

All eight tests passed in Play In Editor, with no crashes or stuck states.

#	Test	Expected result	Result
1	Full round: destroy 3 drones	ROUND COMPLETE appears, timer stops	Pass
2	Press R, play a second round	Score and timer reset, drones return	Pass
3	Play a third full round	No crash, no stuck state	Pass
4	Die three times and respawn	SYSTEM FAILURE, respawn after 3 seconds, full health	Pass
5	Stand on the electric floor	Damage only while it is red	Pass

#	Test	Expected result	Result
6	Walk into the Energy Cell	Health rises by 30, the pickup disappears	Pass
7	Walk to the Control Room door	Opens when close, closes after a delay	Pass
8	Stand behind a wall in front of a drone	The drone does not shoot	Pass

9. Known issues

None of these block gameplay; they are planned for the Beta polish pass.

- **Light warning:** Unreal prints a Virtual Shadow Map "max lights overflow" warning where many local lights overlap. It does not affect gameplay and will be fixed by reducing overlapping lights.
- **Placeholder visuals:** the drones and the Energy Cell are plain spheres, and the arena is still blockout geometry. There is no sound or visual effects yet.
- **Pickup behaviour:** the Energy Cell is used up even when health is full, and it does not respawn.
- **One enemy type:** drones are stationary turrets. They turn and shoot but do not patrol.
- **HUD update method:** score, time and round message use UMG property bindings, which are checked every frame. This is fine at this scale and could become event-driven later.

10. Next steps and AI assistance

Beta will focus on polish and a second interactive mechanic, and Final on content, balance and packaging.

Milestone	Due	Planned work
Beta	Dec 4, 2026	Replace placeholder meshes with final assets; level dressing (cover, emissive trim, props); sound and visual effects (shots, hits, pickups, drone destruction); a second interactive mechanic (elevator or teleporter pad); pickup respawn and a "do not use at full health" check
Final	Dec 18, 2026	Patrol movement for drones using a NavMesh; a second drone type; a second hazard (falling debris); an Overcharge damage pickup; difficulty balancing; bug fixing and a packaged build